Hints: Abstract algebra – 2. Finite rings and division rings

Class work

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

(a)

Hint 1. In a finite ring, s_a is injective iff it is surjective, i.e. iff $ab = 1$ for some b. So it is enough to show that a one-sided inverse in a finite ring is two-sided.

Hint 2. If $ab = 1$, then s_b is injective ($bx = 0 \Rightarrow x = abx = 0$), hence surjective, so $bc = 1$ for some c. Then $a = a(bc) = (ab)c = c$. So a is a unit and d_a is injective too. The converse is symmetric.

(b)

Hint 1. The ring must be infinite. Take the endomorphisms of the space of real sequences, with the right shift $S(x_1, x_2, \dots) = (0, x_1, x_2, \dots)$ and the left shift T, so that $TS = 1 \neq ST$.

Hint 2. For $b = S$: s_S is injective because S is injective, but d_S is not, since $(1 - ST)S = S - S(TS) = 0$ while $1 - ST \neq 0$.

Problem 2

Hint 1. The hypothesis says $ab^2 = b$. Show by induction that $ab^{k+1} = b^k$, and hence $a^k b^{k+1} = b$, for all $k \geqslant 1$. Since A is finite, the powers of b are eventually periodic: there is $N \geqslant 2$ with $b^{2N} = b^N$.

Hint 2. Show $b = b^{N+1}$: $b = a^N b^{N+1} = a^N b^{2N+1} = (a^N b^{N+1}) b^N = b \cdot b^N$. So $e = b^N$ is an idempotent with $eb = be = b$.

Hint 3. Multiply $ab^2 = b$ on the right by b^{2N-2} to get $ae = b^{2N-1}$, so $ab = aeb = b^{2N} = e$. Then $b(ab - 1) = be - b = 0$.

Problem 3

Hint 1. Expand $(x + x)^2 = x + x$ and $(x + y)^2 = x + y$, as in the theoretical minimum.

Hint 2. Characteristic 2 makes $(A, +)$ an $\mathbb{F}_2$-vector space, so a finite Boolean ring has 2^k elements. Conversely, $\mathbb{F}_2^k$ with coordinatewise operations is Boolean, and you can transport its structure to any set with 2^k elements along a bijection.

Hint 3. For $\mathbb{N}_0$, find a *countably infinite* Boolean ring with unit and transport its structure along a bijection. The finite subsets of $\mathbb{N}$ under $\triangle, \cap$ have no unit, but the finite-or-cofinite subsets of $\mathbb{N}$ do (the unit is $\mathbb{N}$).

Problem 4

Note. For a non-commutative ring the map $x \mapsto x^p$ need not be additive: in $M_2(\mathbb{F}_2)$, $(E_{12}+E_{21})^2 = I \neq 0 = E_{12}^2 + E_{21}^2$. The hints assume A is commutative.

Hint 1. $p \cdot 1 = 0$ makes $(A, +)$ an $\mathbb{F}_p$-vector space, so $|A| = p^k$. Additivity of $x \mapsto x^p$: expand by the binomial theorem and use $p \mid \binom{p}{i}$ for $0 < i < p$.

Hint 2. The nilpotent elements form an additive subgroup N. For closure under addition, use the binomial theorem or the Frobenius: $(x+y)^{p^m} = x^{p^m} + y^{p^m}$. Also $1 \notin N$, so N is a proper subgroup of $(A, +)$.

Hint 3. By Lagrange, $|N|$ divides p^k and is less than p^k, so $|N| \leqslant p^{k-1} = |A|/p$.

Problem 5

Hint 1. For each a, the sequence $a, a^2, a^3, \ldots$ takes values in a finite set, so it is eventually periodic: $a^{p_a+t_a} = a^{p_a}$ for some $p_a \geqslant 1$, $t_a \geqslant 1$. Then $a^{q+t} = a^q$ for all $q \geqslant p_a$ and all multiples t of t_a.

Hint 2. Take $p = \max_a p_a$, $t = \operatorname{lcm}_a t_a$ and $m = p + t$.